

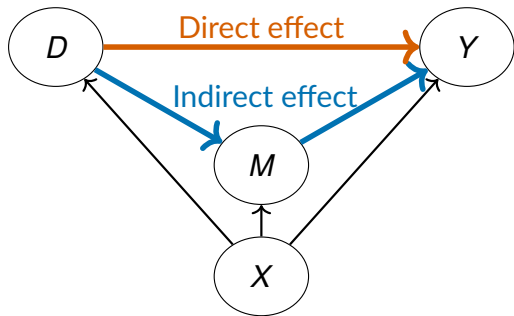
Testing Full Mediation of Treatment Effects and the Identifiability of Causal Mechanisms

Martin Huber Kevin Kloiber Lukáš Lafférs

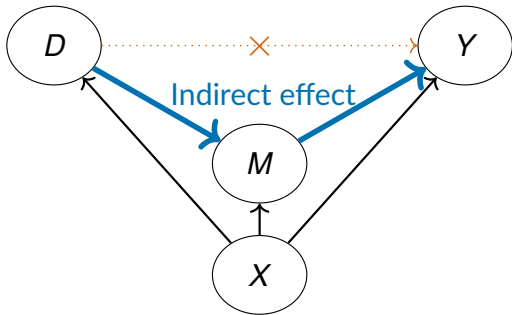
Uni Fribourg · Uni Copenhagen · IER SAV / NHH

SEAM · NOeG · Vienna 2026

Decomposition of effects



Full mediation condition



Why mechanisms matter

- ▶ scale-up: replicate by targeting M
- ▶ policy design: intervene on the right thing
- ▶ theory testing: confirm the hypothesised channel

The identification challenge

- ▶ M is rarely randomised
- ▶ unobserved M - Y confounders bias estimates
- ▶ hard to test even with randomised D

Why testing full mediation is hard

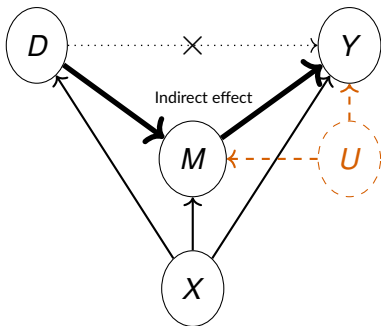
Naive approach: regress Y on D and M ,
test $\hat{\beta}_D \approx 0$

Problem: if M is **endogenous**,
conditioning on it opens a collider
($D \rightarrow M \leftarrow U \rightarrow Y$)

Bias arises even when D is randomized

M is a textbook *bad control*

► Algebra



Mediation questions you know

Setting	Treatment → mediator → outcome	Familiar tool
ALMP	training → re-employment → earnings	LATE, Heckman selection
Social assistance	reform → take-up → household income	take-up models
Migration	inflow → labour supply → native wages	shift-share IV
Returns to schooling	education → occupation / skills → wages	Mincer decomposition
Wage gap decomp.	group → characteristics → wage	Oaxaca-Blinder
RCT analysis	treatment arm → behaviour → outcome	IV, control function

Existing tools need a functional form (Oaxaca-Blinder) or an instrument for M .
We ask a smaller question: *is the full-mediation story consistent with the data?*

Literature

- ▶ mediation analysis
Robins and Greenland (1992), Robins (2003), Pearl (2001), Tchetgen Tchetgen and Shpitser (2012)
- ▶ problems
Acharya, Blackwell, and Sen (2016)
- ▶ testing
Huber and Kueck (2022)
- ▶ full mediation
Kwon and Roth (2026) [▶ More](#)

How it sits next to IV and sensitivity analysis

[▶ Map](#)

Contribution

- ▶ joint test for full mediation and mediator exogeneity
- ▶ clarify the relationship between assumptions
- ▶ DML implementation, simulations, applications

The test needs no instrument for M .

Identification

The key **testable implication (TI)**

$$Y \perp\!\!\!\perp D \mid M, X$$

Outcome independent of treatment,
conditional on mediator and covariates

What is the test really doing?

Given the mediator M and covariates X , **does D still help predict Y ?**

Reject (TI)

- ▶ a direct path $D \rightarrow Y$, **or**
- ▶ M is endogenous, **or both**

Job training. Rejection: knowing re-employment and X , the training arm still predicts long-run earnings.

Do not reject (TI)

- ▶ consistent with both holding
- ▶ not proof of either (power)

Theorem 1 (randomly assigned treatment)

If treatment is randomly assigned and it does move the mediator:

$$\text{full mediation and exogenous mediator} \iff Y \perp\!\!\!\perp D \mid M, X$$

In assumption numbers: under 1, 3, 4 we have $5, 6 \iff \text{(TI)}$ [► Assumptions](#)

- One test covers **both** conditions at once
- Rejection means at least one of them fails
- Proved over all graphs on (X, D, M, Y) with pairwise confounders

Theorem 2 (observational data)

Without randomisation, if treatment still moves the mediator:

no direct D - Y confounding, full mediation, exogenous mediator

$$\Longleftrightarrow Y \perp\!\!\!\perp D \mid M, X$$

In assumption numbers: under 1, 3 we have 4a, 5, 6 $\Longleftrightarrow$ (TI) [Assumptions](#) [Front-door](#)

- ▶ Both stay testable without an RCT
- ▶ Treatment-mediator confounding (4b) is **not** detected by the test
- ▶ Non-rejection alone does not identify the indirect effect

Using the test in practice

What you need

- ▶ D binary treatment
- ▶ M the claimed mediator
- ▶ X covariates, can be many

On sample size. A moderate direct effect ($\gamma = 0.2$) is caught about half the time at $n = 1,000$, almost always at $n = 4,000$. In small samples a non-rejection is weak evidence.

What it will not do

- ▶ tell you *which* assumption failed
- ▶ detect D - M confounding
- ▶ give the size of the direct effect

Estimation ▶ DML · Simulations ▶ Results

Empirical Illustrations

Social norms (Bursztyn et al. 2020, Saudi Arabia)

D Information on peers' attitudes

M Job-matching service sign-up

Y Wife applied for outside job

Controls: baseline beliefs, employment,
education, demographics · $n = 284$

Mediator	<i>p</i> -value
Job-matching	0.004 **
Kwon & Roth (2026): $p = 0.020$	

The stronger rejection may also reflect mediator endogeneity

► More

Perinatal depression (Baranov et al. 2020, Pakistan)

D CBT intervention (Thinking Healthy)

M Grandmother presence;
husband relationship quality

Y Financial empowerment (7 yrs later)

Controls: age, education, employment,
depression severity, wealth, family structure,
... · $n \sim 600$

Mediator	<i>p</i> -value
Grandmother	0.011 *
Relationship	0.022 *
Both	0.013 *
Kwon & Roth (2026): jointly $p = 0.654$	

*A mechanism is missing, or
mediators are endogenous*

► More

Summary

$Y \perp\!\!\!\perp D \mid M, X$ jointly tests **full mediation** and **mediator exogeneity**

- ▶ **Randomized D :** full mediation *and* identifiability jointly testable
- ▶ **Observational D :** full mediation still testable; D-M confounding is not
- ▶ **DML:** $\sqrt{n}$ -consistent, doubly robust, high-dimensional X
- ▶ Both applications **reject** the joint null

Is there a specific channel through which the treatment works dominant?

TI can be used to test it.

Thank you for your attention.

`lukaslaffers.github.io`

$$H_0 : E[Y \mid M, X, D=1] - E[Y \mid M, X, D=0] = 0$$

- ▶ Estimated by **double machine learning**
- ▶ Lasso or random forests for the nuisances, so X can be high-dimensional
- ▶ Doubly robust, Neyman-orthogonal, 5-fold cross-fitting
- ▶ $\sqrt{n}$ -consistent and asymptotically normal

Results (rejection rates at the 5% level)

Scenario		$n = 1,000$	$n = 4,000$
<i>Main design, $\lambda = 0$</i>			
Null	$\delta = 0, \gamma = 0$	0.055	0.040
M-Y conf.	$\delta = 0.25, \gamma = 0$	0.969	1.000
Direct effect	$\delta = 0, \gamma = 0.2$	0.532	1.000
<i>With D-M confounding, $\lambda = 0.25$</i>			
Null	$\delta = 0, \gamma = 0$	0.030	0.037
M-Y conf.	$\delta = 1, \gamma = 0$	1.000	1.000
Direct effect	$\delta = 0, \gamma = 0.2$	0.482	0.993

Size is unaffected by D-M confounding · power is limited at $n = 1,000$

Simulation design

$$D = \mathbf{1}\{X'\beta + U_1 + \lambda U_2 > 0\} \quad \leftarrow D\text{-}M \text{ confounding if } \lambda > 0$$

$$M = 0.5D + X'\beta + \delta U_1 + U_2$$

$$Y = M + X'\beta + \gamma D + \delta U_1 + U_3$$

	Parameters	Violation tested
Null holds	$\delta = 0, \gamma = 0$	none
Null fails	$\delta \neq 0$	$D\text{-}M\text{-}Y$ confounding
Null fails	$\gamma \neq 0$	direct effect $D \rightarrow Y$

$p = 200$ covariates · $n \in \{1000, 4000\}$ · 1000 replications · lasso, 5-fold cross-fitting

Theorem 1: $\lambda = 0$ · Theorem 2: $\lambda = 0.25$ (introduces $D\text{-}M$ confounding)

Setup

Observed variables

D	treatment $\{0, 1\}$
M	mediator
Y	outcome
X	covariates

Potential outcomes

$M(d)$	mediator under d
$Y(d, m)$	outcome under (d, m)
$Y(m)$	under full mediation

Assumptions

1	causal structure	excl. restrictions
2	common support	$p(D, M X) \in (0, 1)$
3	first stage	$\Pr(M(1) \neq M(0)) > 0$
4	random treatment	$D \perp\!\!\!\perp Y(d, m), M(d) \mid X$
4a	exog. treatment (wrt Y)	$D \perp\!\!\!\perp Y(d, m) \mid X$
4b	exog. treatment (wrt M)	$D \perp\!\!\!\perp M(d) \mid X$
5	cond. full mediation	$Y(d, m) = Y(m) \mid X$ a.s.
6	exog. mediator	$M \perp\!\!\!\perp Y(d, m) \mid D, X$

► Back to Theorem 1

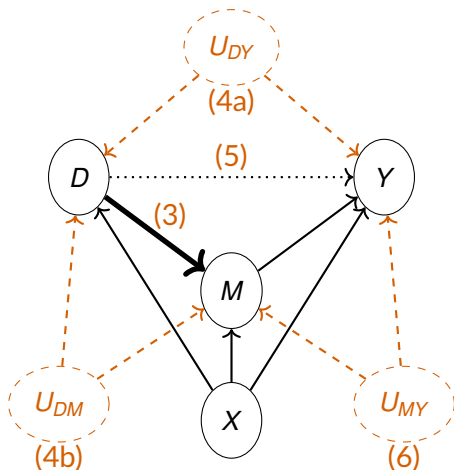
Identifying Assumptions

Assumption 1

- ▶ $M(y) = M$:
($Y \not\rightarrow M$, Y does not cause M)
- ▶ $D(m, y) = D$
- ▶ $X(d, m, y) = X$
- ▶ Faithfulness

Assumption 3

- ▶ $D \rightarrow M$ exists



Relation to front-door / back-door

$$\underbrace{\Pr(Y=y \mid D=d, X)}_{\text{back-door}} = \underbrace{\sum_m \Pr(M=m \mid D=d, X) \sum_{d'} \Pr(Y=y \mid D=d', M=m, X) \Pr(D=d' \mid X)}_{\text{front-door}} \quad (\text{BD=FD})$$

Theorem 3. $(\text{TI}) \Rightarrow (\text{BD} = \text{FD})$

Theorem 4. $(\text{BD} = \text{FD}) + \text{Separability} \Rightarrow (\text{TI})$

BD-FD equivalence is a *weaker* test than (TI)

Separability: $\Pr(Y=y \mid D, M, X) = \alpha(D, X) + \beta(M, X)$
rules out D - M interactions; strong assumption

Double Machine Learning (score)

$$H_0 : E[Y \mid M, X, D=1] - E[Y \mid M, X, D=0] = 0$$

Doubly robust score (Apfel et al. 2023; Chernozhukov et al. 2018)

$$\begin{aligned}\tilde{\psi} = & (\mu_1 - \mu_0)^2 + 2(\mu_1 - \mu_0) \left[\frac{(Y - \mu_1)D}{p} - \frac{(Y - \mu_0)(1 - D)}{1 - p} \right] \\ & + (\mu_1 - \mu_0) + \left[\frac{(Y - \mu_1)D}{p} - \frac{(Y - \mu_0)(1 - D)}{1 - p} \right] - \theta\end{aligned}$$

$$\mu_d = E[Y \mid M, X, D=d], \quad p = \Pr(D=1 \mid M, X)$$

- ▶ **Neyman orthogonality:** insensitive to nuisance estimation error
- ▶ **5-fold cross-fitting:** avoids overfitting bias
- ▶ **Lasso / ML nuisances:** $\sqrt{n}$ -consistent, asymptotically normal

The bad-control problem, in algebra

Linear DGP with full mediation *but* unobserved M - Y confounder U :

$$M = \alpha D + U + \varepsilon_M$$

$$Y = \beta M + \delta U + \varepsilon_Y \quad (\text{no } D \text{ in } Y)$$

Run the **naïve** Baron-Kenny regression Y on D, M :

$$\hat{\beta}_D \xrightarrow{p} -\frac{\alpha \delta \sigma_U^2}{\sigma_M^2} \neq 0$$

Even with randomised D , conditioning on M opens the collider.

Naive test concludes: *direct effect exists.*

Truth: full mediation holds; M is endogenous.

► Back

Angrist-Pischke would say...

- M is a **bad control**: post-treatment, endogenous
- Bias depends on δ and σ_U^2 , not on the true structure
- No reason to think $\hat{\beta}_D \approx 0$ implies full mediation

Our test: rejection comes from a direct effect *or* M -endogeneity; non-rejection rules out both.

How this fits with other approaches

Approach	Key requirement	What it delivers
IV for M	valid IV $Z \rightarrow M$	point ID of indirect effect
Control function	model of M + error distribution	point ID, under restrictions
Sensitivity (Imai–Keele)	bound on M – Y confounding	bounds on indirect effect
Front-door (Pearl)	full mediation <i>assumed</i>	point ID of the effects
Partial ID (Kwon–Roth)	monotonicity, finite M	sharp bounds and test
Our test (TI)	randomised D + Asm. 1, 3	joint test of both; DML, high-dim X

Complementary, not competing. Run the test first. If you reject, IV or sensitivity analysis tells you how far to trust an indirect-effect estimate.

Comparison to Kwon and Roth (2026)

Kwon and Roth (2026):

- ▶ tests sharp null of full mediation
- ▶ partial identification approach
- ▶ sharp bounds via linear program
- ▶ discretization of cont. variables, M has finite support
- ▶ monotonicity assumptions
- ▶ degree of full mediation violations
- ▶ provide several modifications (e.g. violations of monotonicity)
- ▶ many covariates X

Kwon, S. and Roth, J. (2026). Testing mechanisms. forthcoming in *The Review of Economic Studies*

Why mean independence?

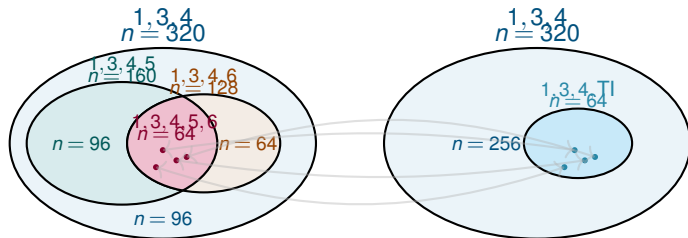
$$H_0 : E[Y \mid M, X, D=1] - E[Y \mid M, X, D=0] = 0$$

Why mean independence instead of full independence?

- ▶ **ATE:** mean independence suffices
- ▶ **full CI:** needs non-parametric distributional tests
- ▶ **mean independence:** allows doubly robust DML

▶ [Back to Testing](#)

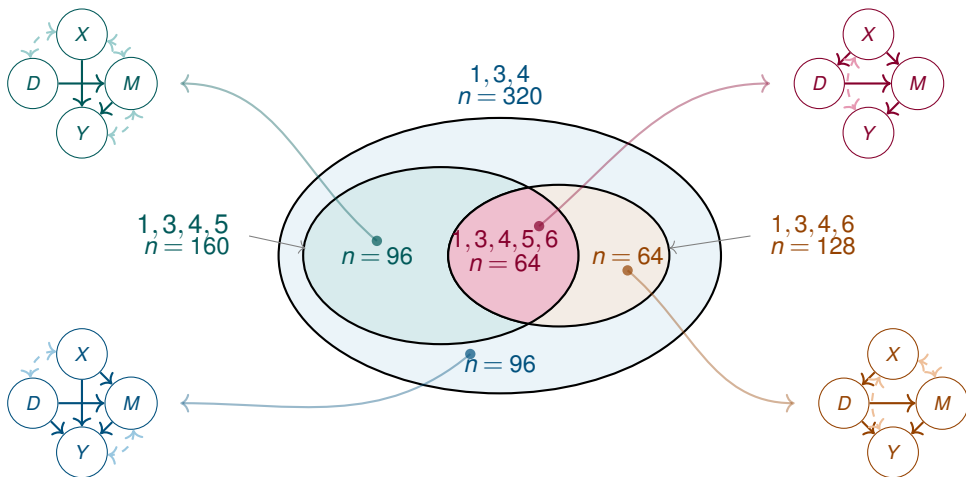
Theorem 1 (graph classes)



Under Assumptions 1, 3, 4 :

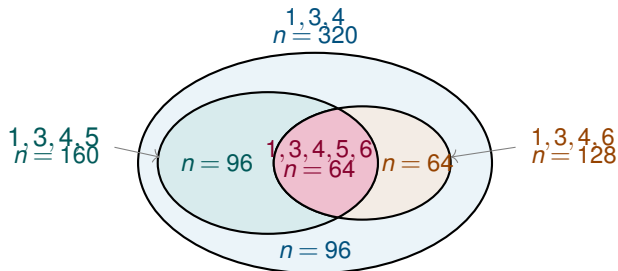


Theorem 1 (visual)



Graph search (counting)

- ▶ 34752 DAGs with obs. variables (X, D, M, Y) and pairwise confounders
- ▶ 4096 that satisfy Assumption 1
- ▶ 2048 that satisfy Assumption 1 and 3
- ▶ 320 that satisfy Assumption 1, 3 and 4
- ▶ 64 that satisfy Assumption 1, 3, 4, 5 and 6



- ▶ men underestimate how open are other men about women working outside of the home
- ▶ experiment: receive info on other men's beliefs or not
- ▶ job-search service or gift card
- ▶ treated men's women prob of applying for job-interview higher within 6 months
- ▶ is this short-run sign-up for job-search service the main driver of the outcomes in the longer-term?
- ▶ $n = 284$

Bursztyn, L., Gonzalez, A. L., and Yanagizawa-Drott, D. (2020). Misperceived social norms: Women working outside the home in Saudi Arabia. *American Economic Review*, 110(10), 2997–3029.

- ▶ RCT of CBT to reduce depression for pregnant women in rural Pakistan
- ▶ studied effect on financial empowerment
- ▶ grandmother presence (proxy for family support) and relationship with husband as mediators
- ▶ $n \sim 600$

Baranov, V., Bhalotra, S., Biroli, P., and Maselko, J. (2020). Maternal depression, women's empowerment, and parental investment: Evidence from a randomized controlled trial. *American Economic Review*, 110(3), 824–859.

References

- ▶ Acharya, A., Blackwell, M., and Sen, M. (2016). Explaining causal findings without bias: Detecting and assessing direct effects. *American Political Science Review*, 110, 512–529.
- ▶ Apfel, N., Hatamyar, J., Huber, M., and Kueck, J. (2023). Learning control variables and instruments for causal analysis in observational data. Working paper, University of Fribourg.
- ▶ Baranov, V., Bhalotra, S., Biroli, P., and Maselko, J. (2020). Maternal depression, women's empowerment, and parental investment: Evidence from a randomized controlled trial. *American Economic Review*, 110(3), 824–859.
- ▶ Bursztyn, L., Gonzalez, A. L., and Yanagizawa-Drott, D. (2020). Misperceived social norms: Women working outside the home in Saudi Arabia. *American Economic Review*, 110(10), 2997–3029.
- ▶ Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68.
- ▶ Huber, M. and Kueck, J. (2022). Testing the identification of causal effects in observational data. arXiv preprint arXiv:2203.15890.
- ▶ Kwon, S. and Roth, J. (2026). Testing mechanisms. forthcoming in *The Review of Economic Studies*
- ▶ Pearl, J. (2001). Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, 411–420. Morgan Kaufmann, San Francisco.
- ▶ Robins, J. M. (2003). Semantics of causal DAG models and the identification of direct and indirect effects. In P. J. Green, N. L. Hjort, and S. Richardson (Eds.), *Highly Structured Stochastic Systems*, 70–81. Oxford University Press.
- ▶ Robins, J. M. and Greenland, S. (1992). Identifiability and exchangeability for direct and indirect effects. *Epidemiology*, 3, 143–155.
- ▶ Tchetgen, E. J. T., & Shpitser, I. (2012). Semiparametric theory for causal mediation analysis: efficiency bounds, multiple robustness, and sensitivity analysis. *Annals of statistics*, 40(3), 1816.